

TECHSHIP INTERNSHIP

Day 2- JavaScript Basics

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Today's session focused on learning the fundamentals of JavaScript. I explored concepts such as variables, data types, operators, conditional statements, and loops. Through practical exercises, I created simple programs including a student information system, grade calculator, attendance calculator, multiplication table generator, and a student result management system.

In addition to coding, I learned how to organize project folders, create separate files for different exercises, and run JavaScript programs using Node.js in the VS Code terminal. Understanding file paths and executing files from the correct directory was initially challenging, but practicing these tasks helped me become more comfortable with the development workflow.

The most interesting part of today's learning was using conditional statements to make decisions and loops to automate repetitive tasks. Overall, the session strengthened my understanding of JavaScript basics and increased my confidence in writing and executing simple programs independently.

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the project structure: TECHSHIP > Day1 > day1.js, README.md, TECHSHIP Day-1.pdf, and Day2 > Exercise > Ex1.js. The main editor window shows the code for Ex1.js, which defines variables for a student's name, age, college, course, and year, and then logs this information to the console. The terminal at the bottom shows the command prompt where the file is executed, resulting in the printed student details.

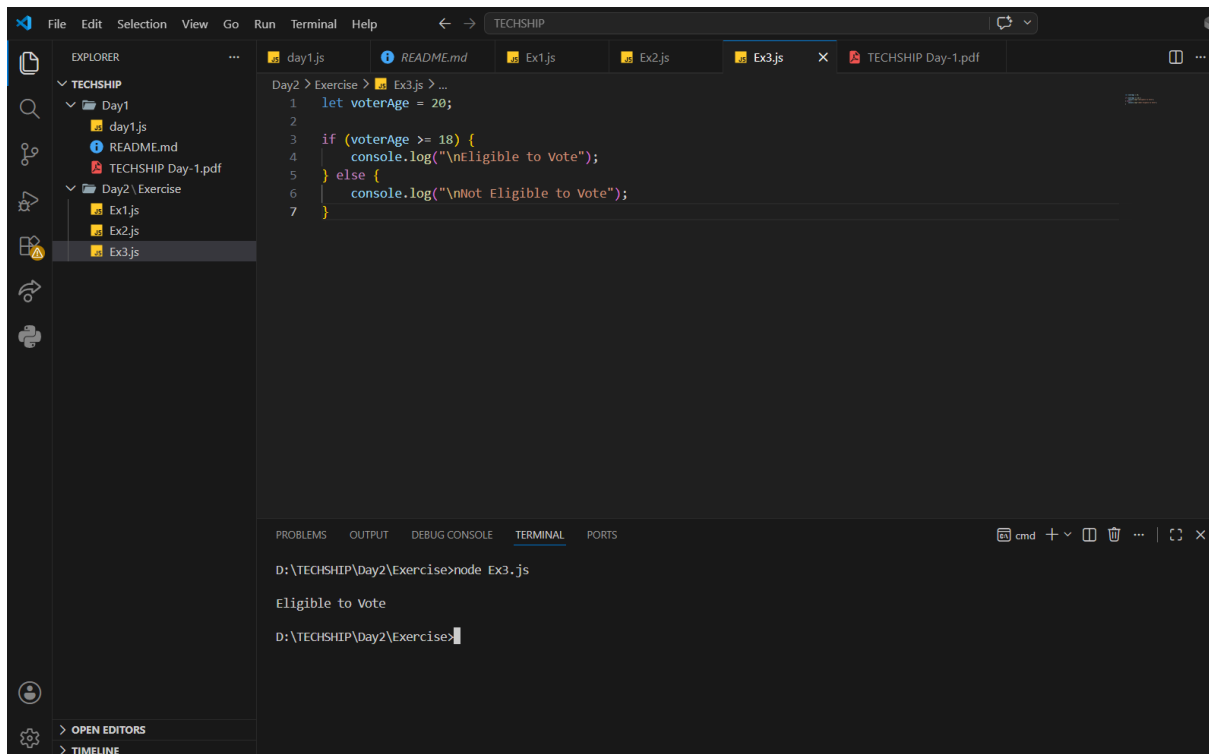
```
File Edit Selection View Go Run Terminal Help < -> TECHSHIP
EXPLORER
TECHSHIP
  Day1
    day1.js
    README.md
    TECHSHIP Day-1.pdf
  Day2\Exercise
    Ex1.js
day1.js README.md Ex1.js x TECHSHIP Day-1.pdf
Day2 > Exercise > Ex1.js > name
1 let name = "Kelna Sebastian";
2 let age = 21;
3 let college = "AISAT";
4 let course = "Btech";
5 let year = 3;
6
7 console.log("Student Information");
8 console.log("Name:", name);
9 console.log("Age:", age);
10 console.log("College:", college);
11 console.log("Course:", course);
12 console.log("Year:", year);
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
D:\TECHSHIP\Day2>cd Exercise
D:\TECHSHIP\Day2\Exercise>node Ex1.js
Student Information
Name: Kelna Sebastian
Age: 21
College: AISAT
Course: Btech
Year: 3
D:\TECHSHIP\Day2\Exercise>
```

Ex1:Student Information System

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the project structure: TECHSHIP > Day1 > day1.js, README.md, TECHSHIP Day-1.pdf, and Day2 > Exercise > Ex1.js, Ex2.js. The main editor window shows the code for Ex2.js, which calculates the total marks, average marks, and percentage for a student based on five subject marks. The terminal at the bottom shows the command prompt where the file is executed, resulting in the printed student grade report.

```
File Edit Selection View Go Run Terminal Help < -> TECHSHIP
EXPLORER
TECHSHIP
  Day1
    day1.js
    README.md
    TECHSHIP Day-1.pdf
  Day2\Exercise
    Ex1.js
    Ex2.js
day1.js README.md Ex1.js Ex2.js x TECHSHIP Day-1.pdf
Day2 > Exercise > Ex2.js > ...
1 let sub1 = 85;
2 let sub2 = 90;
3 let sub3 = 78;
4 let sub4 = 88;
5 let sub5 = 92;
6
7 let total = sub1 + sub2 + sub3 + sub4 + sub5;
8 let average = total / 5;
9 let percentage = average;
10
11 console.log("\nStudent Grade Report");
12 console.log("Total Marks:", total);
13 console.log("Average Marks:", average);
14 console.log("Percentage:", percentage + "%");
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
D:\TECHSHIP\Day2\Exercise>node Ex2.js
Student Grade Report
Total Marks: 433
Average Marks: 86.6
Percentage: 86.6%
D:\TECHSHIP\Day2\Exercise>
```

Ex2:Student Grade Calculator



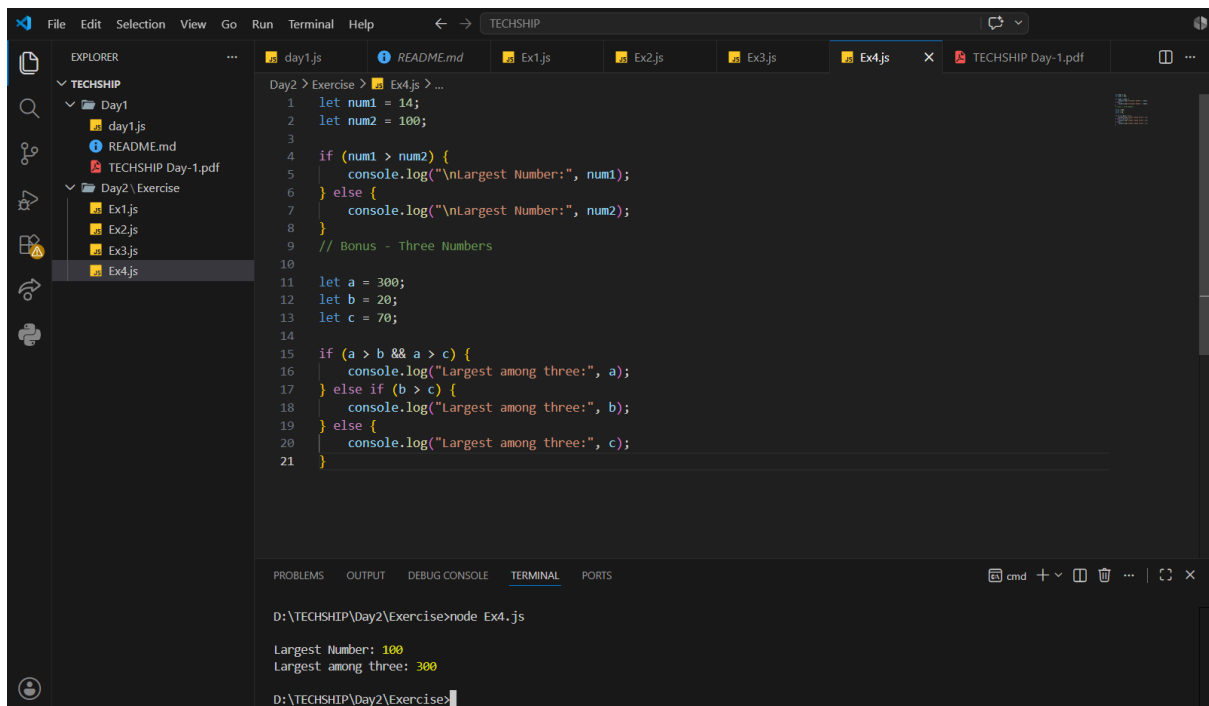
The screenshot shows the Visual Studio Code editor with the file explorer on the left. The 'EXPLORER' pane shows a project named 'TECHSHIP' with a folder 'Day2' containing 'Exercise' and 'Ex3.js'. The 'EXERCISE' pane shows the code for 'Ex3.js'.

```
Day2 > Exercise > Ex3.js > ...
1 let voterAge = 20;
2
3 if (voterAge >= 18) {
4   console.log("\nEligible to Vote");
5 } else {
6   console.log("\nNot Eligible to Vote");
7 }
```

The 'TERMINAL' pane at the bottom shows the command prompt output:

```
D:\TECHSHIP\Day2\Exercise>node Ex3.js
Eligible to Vote
D:\TECHSHIP\Day2\Exercise>
```

Ex3: Voting Eligibility Checker



The screenshot shows the Visual Studio Code editor with the file explorer on the left. The 'EXPLORER' pane shows a project named 'TECHSHIP' with a folder 'Day2' containing 'Exercise' and 'Ex4.js'. The 'EXERCISE' pane shows the code for 'Ex4.js'.

```
Day2 > Exercise > Ex4.js > ...
1 let num1 = 14;
2 let num2 = 100;
3
4 if (num1 > num2) {
5   console.log("\nLargest Number:", num1);
6 } else {
7   console.log("\nLargest Number:", num2);
8 }
9 // Bonus - Three Numbers
10
11 let a = 300;
12 let b = 20;
13 let c = 70;
14
15 if (a > b && a > c) {
16   console.log("Largest among three:", a);
17 } else if (b > c) {
18   console.log("Largest among three:", b);
19 } else {
20   console.log("Largest among three:", c);
21 }
```

The 'TERMINAL' pane at the bottom shows the command prompt output:

```
D:\TECHSHIP\Day2\Exercise>node Ex4.js
Largest Number: 100
Largest among three: 300
D:\TECHSHIP\Day2\Exercise>
```

Ex4: Largest Number Finder

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the file structure of a project named 'TECHSHIP'. The file 'Ex5.js' is selected under the 'Day2 > Exercise' folder. The main editor window shows the code for 'Ex5.js', which calculates the attendance percentage based on the number of attended classes and total classes. The code is as follows:

```
Day2 > Exercise > Ex5.js > attendedClasses
1 let totalClasses = 63;
2 let attendedClasses = 57;
3
4 let attendancePercentage =
5   (attendedClasses / totalClasses) * 100;
6
7 console.log(
8   "\nAttendance Percentage:",
9   attendancePercentage + "%"
10  );
```

The bottom panel shows the 'TERMINAL' tab with the command prompt output:

```
D:\TECHSHIP\Day2\Exercise>node Ex5.js
Attendance Percentage: 90.47619047619048%
D:\TECHSHIP\Day2\Exercise>
```

Ex5:Attendance Calculator

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the file structure of a project named 'TECHSHIP'. The file 'Ex6.js' is selected under the 'Day2 > Exercise' folder. The main editor window shows the code for 'Ex6.js', which generates a multiplication table for a given number. The code is as follows:

```
Day2 > Exercise > Ex6.js > ...
1 let number = 13;
2
3 console.log("\nMultiplication Table of", number);
4
5 for (let i = 1; i <= 10; i++) {
6   console.log(
7     number + " x " + i + " = " + (number * i)
8   );
9 }
```

The bottom panel shows the 'TERMINAL' tab with the command prompt output:

```
D:\TECHSHIP\Day2\Exercise>node Ex6.js
Multiplication Table of 13
13 x 1 = 13
13 x 2 = 26
13 x 3 = 39
13 x 4 = 52
13 x 5 = 65
13 x 6 = 78
13 x 7 = 91
13 x 8 = 104
13 x 9 = 117
13 x 10 = 130
D:\TECHSHIP\Day2\Exercise>
```

Ex6:Multiplication Table Generator

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
  Day1
    day1.js
    README.md
    TECHSHIP Day-1.pdf
  Day2\Exercise
    Ex1.js
    Ex2.js
    Ex3.js
    Ex4.js
    Ex5.js
    Ex6.js
    Ex7.js

Day2 > Exercise > Ex7.js > ...
1 let sum = 0;
2
3 for (let i = 1; i <= 100; i++) {
4   sum += i;
5 }
6
7 console.log("\nSum of numbers from 1 to 100 =", sum);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
D:\TECHSHIP\Day2\Exercise>node Ex7.js

Sum of numbers from 1 to 100 = 5050

D:\TECHSHIP\Day2\Exercise>
```

Ex7:Sum of Natural Numbers

Challenge Problems

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
  Day1
    day1.js
    README.md
    TECHSHIP Day-1.pdf
  Day2
    Challenges
      Ch1.js
    Exercise

Day2 > Challenges > Ch1.js > ...
1 let evenNumbers = "";
2 let oddNumbers = "";
3
4 for (let i = 1; i <= 100; i++) {
5   if (i % 2 === 0) {
6     evenNumbers += i + " ";
7   } else {
8     oddNumbers += i + " ";
9   }
10 }
11
12 console.log("Even Numbers:", evenNumbers);
13 console.log("Odd Numbers :", oddNumbers);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
D:\TECHSHIP\Day2\Challenges>node Ch1.js

Even Numbers: 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100
Odd Numbers : 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99

D:\TECHSHIP\Day2\Challenges>
```

1.Even and Odd Number Generator

The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows the file structure: 'TECHSHIP' > 'Day1' > 'day1.js', 'README.md', 'TECHSHIP Day-1.pdf', 'Day2' > 'Challenges' > 'Ch1.js', 'Ch2.js' (selected), and 'Exercise'. The main editor window displays the code for 'Ch2.js':

```
Day2 > Challenges > Ch2.js > ...
1  let output = "";
2
3  for (let i = 1; i <= 50; i++) {
4
5      if (i % 3 === 0 && i % 5 === 0) {
6          output += "FizzBuzz ";
7      }
8      else if (i % 3 === 0) {
9          output += "Fizz ";
10     }
11     else if (i % 5 === 0) {
12         output += "Buzz ";
13     }
14     else {
15         output += i + " ";
16     }
17 }
18
19 console.log(output);
```

Below the code editor is the TERMINAL panel, which shows the output of running 'node Ch2.js':

```
D:\TECHSHIP\Day2\Challenges>node Ch2.js
1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz 16 17 Fizz 19 Buzz Fizz 22 23 Fizz Buzz 26 Fizz 28 29 FizzBuzz 31 32 Fizz 34 B
uzz Fizz 37 38 Fizz Buzz 41 Fizz 43 44 FizzBuzz 46 47 Fizz 49 Buzz

D:\TECHSHIP\Day2\Challenges>
```

2.FizzBuzz

The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar shows the file structure, with 'Ch3.js' selected under 'Day2' > 'Challenges'. The main editor window displays the code for 'Ch3.js':

```
Day2 > Challenges > Ch3.js > ...
1  let balance = 10000;
2  let withdrawalAmount = 3000;
3
4  if (withdrawalAmount <= balance) {
5      balance -= withdrawalAmount;
6      console.log("\nWithdrawal Successful");
7      console.log("Remaining Balance:", balance);
8  } else {
9      console.log("\nInsufficient Balance");
10 }
```

Below the code editor is the TERMINAL panel, which shows the output of running 'node Ch3.js':

```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

D:\TECHSHIP>cd Day2/Challenges

D:\TECHSHIP\Day2\Challenges>node Ch3.js

Withdrawal Successful
Remaining Balance: 7000

D:\TECHSHIP\Day2\Challenges>
```

3.ATM Withdrawal System

The screenshot shows a VS Code editor with a file explorer on the left. The file explorer shows a project named 'TECHSHIP' with a folder 'Day2' containing a subfolder 'Challenges'. The 'Challenges' folder contains files 'Ch1.js', 'Ch2.js', 'Ch3.js', and 'Ch4.js'. The 'Ch4.js' file is selected and its code is displayed in the editor. The code is a JavaScript program that calculates an electricity bill based on units consumed, cost per unit, and a fixed charge. The terminal at the bottom shows the command 'D:\TECHSHIP\Day2\Challenges>node Ch4.js' and the output of the program.

```
Day2 > Challenges > Ch4.js > ...
1 let unitsConsumed = 250;
2 let costPerUnit = 8;
3 let fixedCharge = 150;
4
5 let electricityCharge =
6   unitsConsumed * costPerUnit;
7
8 let totalBill =
9   electricityCharge + fixedCharge;
10
11 console.log("\nElectricity Bill");
12 console.log("Units Consumed:", unitsConsumed);
13 console.log("Cost per Unit: ₹" + costPerUnit);
14 console.log("Electricity Charge: ₹" + electricityCharge);
15 console.log("Fixed Charge: ₹" + fixedCharge);
16 console.log("Total Bill: ₹" + totalBill);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\TECHSHIP\Day2\Challenges>node Ch4.js

Electricity Bill
Units Consumed: 250
Cost per Unit: ₹8
Electricity Charge: ₹2000
Fixed Charge: ₹150
Total Bill: ₹2150

D:\TECHSHIP\Day2\Challenges>
```

4.Electricity Bill Calculator

Assignment

The screenshot shows a VS Code editor with a file explorer on the left. The file explorer shows a project named 'TECHSHIP' with a folder 'Day2' containing a subfolder 'Challenges'. The 'Challenges' folder contains files 'Ch1.js', 'Ch2.js', 'Ch3.js', 'Ch4.js', and 'assign.js'. The 'assign.js' file is selected and its code is displayed in the editor. The code is a JavaScript program that calculates a student's average and grade based on their scores in various subjects. The terminal at the bottom shows the command 'D:\TECHSHIP\Day2>node assign.js' and the output of the program.

```
Day2 > assign.js > ...
1 let name="Kelna", roll=101, maths=74, science=93, english=80, computer=95, social=78;
2 let total = maths + science + english + computer + social;
3 let average = total / 5;
4 let percentage=average;
5 let grade;
6 if(average >= 90) grade = "A";
7 else if(average >= 80) grade = "B";
8 else if(average >= 70) grade = "C";
9 else if(average >= 60) grade = "D";
10 else grade = "F";
11 console.log("STUDENT RESULT REPORT");
12 console.log("Name:", name);
13 console.log("Roll No:", roll);
14 console.log("Maths:", maths);
15 console.log("Science:", science);
16 console.log("English:", english);
17 console.log("Computer:", computer);
18 console.log("Social:", social);
19 console.log("Total:", total);
20 console.log("Average:", average);
21 console.log("Percentage", percentage+"%");
22 console.log("Grade:", grade);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\TECHSHIP\Day2>node assign.js
STUDENT RESULT REPORT
Name: Kelna
Roll No: 101
Maths: 74
Science: 93
English: 80
Computer: 95
Social: 78
Total: 420
Average: 84
Percentage 84%
Grade: B
```

Student Result Management System